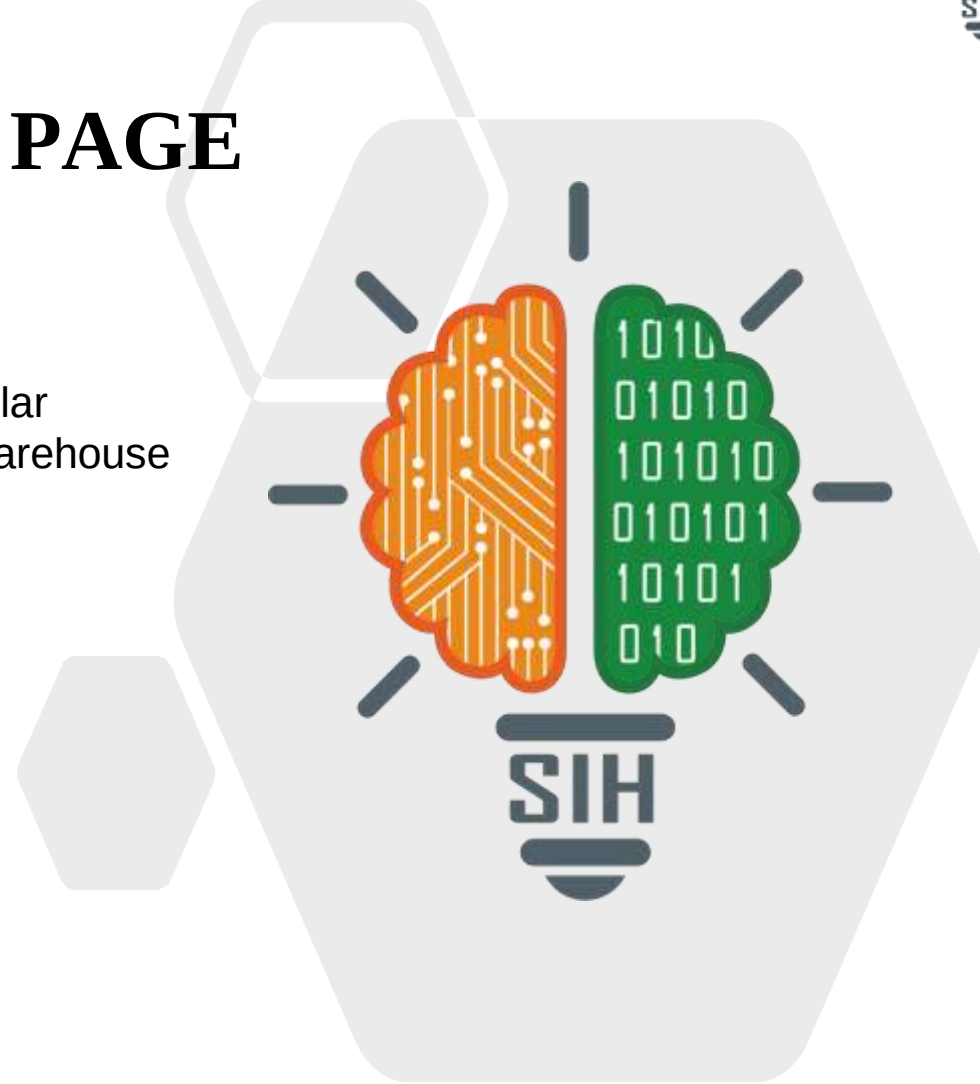


## TITLE PAGE

- **Problem Statement ID** - SIH26112
- **Problem Statement Title** - Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation
- **Theme** - Robotics and Drones
- **PS Category** - Hardware
- **Team ID** - 158819
- **Team Name (Registered on portal)** – Team\_7



# IDEA TITLE

## Autonomous Multi-Level Rack Retrieval Robot for Adaptive Warehouse Automation

### ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

#### The Problem:

- Conventional AMRs primarily transport items at floor level.
- Multi-level racks require a mechanism to access elevated storage locations.
- Fixed AS/RS solutions are typically tied to dedicated rack and aisle infrastructure.
- Different rack heights require flexible vertical positioning during retrieval.
- Existing warehouses need solutions that can integrate without major rack modification.

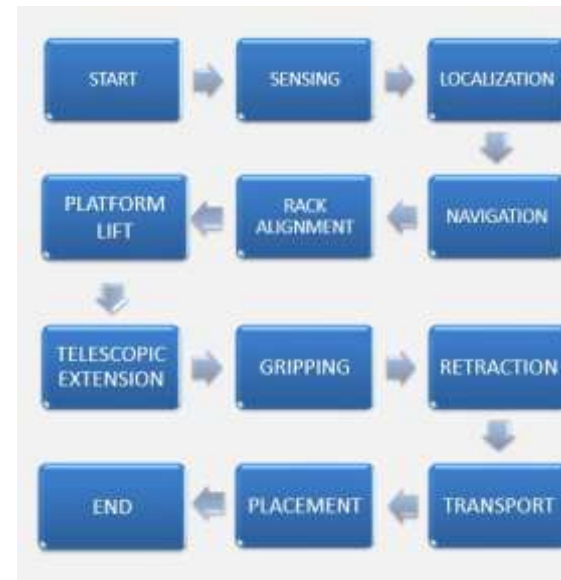
#### Our Solution:

- Autonomous robot for rack item retrieval
- Rising platform reaches the required rack level
- Telescopic arm extends into the rack
- Adjustable gripper accommodates different box sizes
- Designed for existing warehouse racks

#### Key Innovation:

- Mobile + lift + telescopic reach in one robot
- Automated rack-level access across different heights
- Adaptive gripping for varied item/box sizes
- Minimal rack modification for deployment
- Modular mechanical architecture for scalable development

#### How It Works:



# TECHNICAL APPROACH

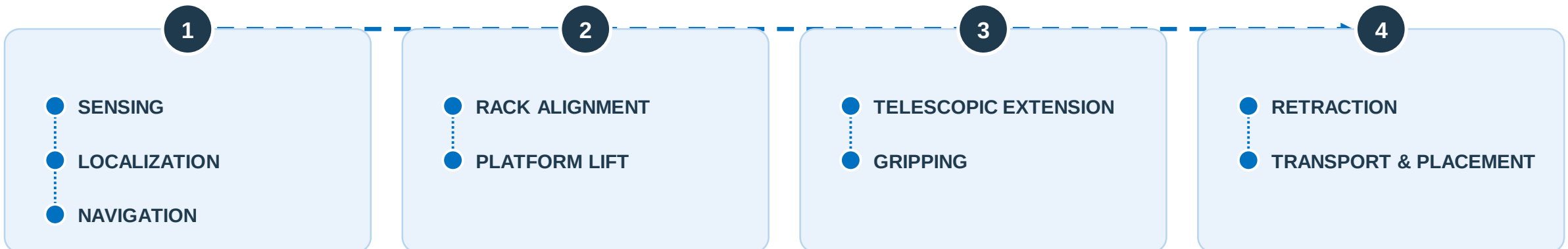
## A. Hardware

|                |                                  |
|----------------|----------------------------------|
| AMR chassis    | Adjustable rubber-padded gripper |
| Motor drivers  | LiDAR / depth sensing            |
| Battery system | IMU & wheel encoders             |
| Camera         | Proximity / safety sensors       |
| Telescopic arm | Raspberry Pi 5 controller        |

## B. Software

|        |                 |
|--------|-----------------|
| ROS 2  | Rviz2           |
| SLAM   | Autodesk Fusion |
| Gazebo | Movelt 2        |
| Nav2   | Codesys         |

## C. System Workflow



# FEASIBILITY AND VIABILITY



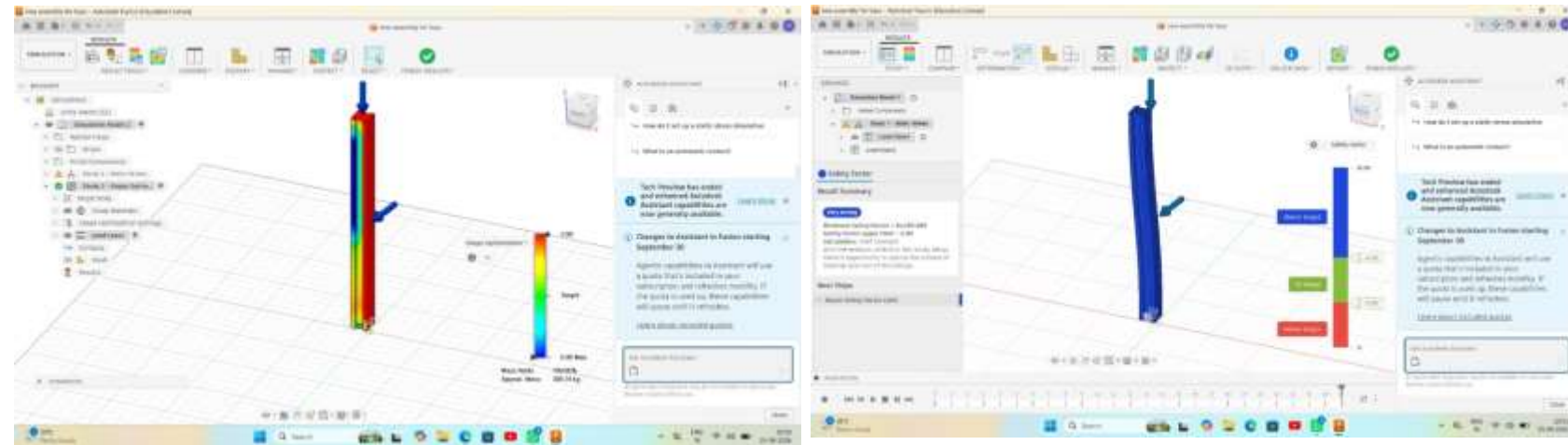
| ASPECT                                 | FEASIBILITY                                      | KEY RISK                              | MITIGATION                                               |
|----------------------------------------|--------------------------------------------------|---------------------------------------|----------------------------------------------------------|
| <b>Navigation &amp; localization</b>   | ROS 2 + LiDAR navigation is well established     | Dynamic obstacles, localization error | Sensor fusion + obstacle detection                       |
| <b>Rack alignment</b>                  | Depth / proximity sensing supports alignment     | Lateral offset at the rack            | Closed-loop fine alignment before lift                   |
| <b>Lifting mechanism</b>               | Actuated linear lifting is readily implementable | Load variation, mechanical deflection | Payload limits + structural analysis + position feedback |
| <b>Telescopic arm &amp; gripper</b>    | Controlled linear reach with adjustable grip     | Cantilever load, varied item sizes    | Payload limits + support / locking + grip range limits   |
| <b>Stability &amp; payload</b>         | Low-mounted battery supports stability           | CG shift during arm extension         | Wide base + low CG + extension limits                    |
| <b>Safety</b>                          | Sensors can support safe operation               | Collision with people or racks        | Obstacle sensing + speed limits + emergency stop         |
| <b>Manufacturing &amp; maintenance</b> | Modular design with standard components          | Wear in moving parts, alignment drift | Modular parts + periodic calibration                     |

Assembly file - <https://a360.co/3Vkj5IU>

For full details of the design with the topology details visit GitHub - <https://github.com/Yuvaraaj-007/SiH-26112/tree/main>

- **Warehouse Operations:** Automated multi-level retrieval; less repetitive manual handling
- **Productivity:** Continuous material movement; consistent retrieval workflow
- **Safety:** Less manual heavy handling; controlled robotic retrieval
- **Infrastructure:** Built for existing racks; minimal rack modification
- **Scalability:** Modular mechanical design; adapts to warehouse layouts
- **Economic Potential:** Potentially less manual handling; reusable multi-task platform

For assembly animation [click here](#)  
For stress analysis video [click here](#)  
For optimization of design [click here](#)



# RESEARCH AND REFERENCES



- 1.ROS 2 — Robotics Middleware** Macenski et al., "Robot Operating System 2: Design, Architecture, and Uses in the Wild," Science Robotics, 2022. <https://doi.org/10.1126/scirobotics.abm6074>
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